SOCIAL MEDIA ANALYTICS

Final Technical Documentation & Project Report

SIH Prototype

Trend Analysis • Sentiment Analysis • Engagement Analytics • Risk Prioritization

August 2026

Executive Summary

This report consolidates the project's technical work on an AI-driven social-media analytics prototype. The work covers dataset preparation, time and trend analysis, engagement analytics, Twitter conversation analysis, sentiment analysis, machine-learning evaluation, and a preliminary risk-prioritization system.

The prototype uses existing CSV datasets. Instagram is used for content, engagement, exposure, timing and audience-growth analysis. Twitter is used for conversation activity, original-versus-retweet analysis and sentiment analysis. A later prototype layer combines sentiment, propagation and engagement signals into a review-priority score.

Important interpretation

The current risk score is a heuristic prioritization signal, not a confirmed threat probability. A negative post is not automatically a threat; multiple signals and human review are required.

Area	Current Result
Instagram dataset	29,999 posts; 23 columns; 366-day observed date span; zero missing values reported
Twitter conversation	Retweets approximately 90% of observed activity
Twitter sentiment	All tweets: 52.94% positive, 38.42% neutral, 8.64% negative
ML sentiment model	TF-IDF + Logistic Regression; 98.74% final accuracy on 398 unseen tweets
Risk prototype	Weighted heuristic using negative sentiment, shares, reach and engagement

1. Project Overview

The initial project strategy was to build a small, reliable working demonstration rather than attempt the complete SIH architecture during the internal round. The broader problem is framed around understanding what people are saying, how they are feeling, who the audience is, and how information spreads.

Analytical Area	Purpose
Trend / Topic Detection	Identify topics, hashtags and frequently discussed terms.
Sentiment & Emotion Analysis	Understand positive, negative and neutral reactions.
Demographic Profiling	Understand audience characteristics.
Network / Link Analysis	Study interactions and information flow.
Timeline Management	Preserve timestamps and analyse changes over time.

1.1 Overall Concept

Core Flow

Collect social-media data → Organise chronologically → Analyse content, sentiment, audience and interactions → Present actionable insights

1.2 Prototype Scope

- Twitter sentiment classification and sentiment-over-time analysis.
- Twitter trend/timeline and basic interaction analysis.
- Instagram content-performance and engagement analysis.
- Simple visual presentation/dashboard of analytical results.

2. Datasets and Data Preparation

2.1 Instagram Dataset

The Instagram dataset contains 29,999 post records and 23 columns/signals. The checked dataset was reported to have zero missing values and an observed post-date span of 366 days.

Data Group	Key Fields
Account	post_id, account_id, account_type
Audience	follower_count
Content	media_type, content_category, caption_length, hashtags_count
Time	post_datetime, post_date, post_hour, day_of_week
Engagement	likes, comments, shares, saves
Exposure	reach, impressions
Growth	followers_gained
Outcome	engagement rate, performance bucket label

2.2 Twitter Dataset

The Twitter dataset contains tweet text, timestamps, users, replies, retweets, likes and retweet counts. It was prepared for timeline, engagement, conversation and sentiment analysis.

- The original dataset contained 17 columns.
- Timestamp information was converted to a proper datetime format.
- Five-minute time bins, hour and minute fields were created.
- is_retweet was standardized and tweet_type was created as Original or Retweet.
- A text_clean field was prepared for sentiment analysis.
- Irrelevant fields were removed while important engagement fields were retained.

3. System Architecture

The prototype architecture is intentionally lightweight. Existing CSV files provide the data, Pandas performs cleaning and structuring, separate analysis streams process Instagram and Twitter, and the outputs are presented through visual summaries.

Stage	Technology / Source	Role
Data Source	Existing CSV files	Provide prototype data; no live API or streaming required at this stage.
Processing	Pandas	Load, clean, structure and normalize data.
Twitter Stream	Twitter dataset	Sentiment, trends/timeline and basic network analysis.
Instagram Stream	Instagram dataset	Content, engagement, reach, timing and growth analysis.
Output	Dashboard / visual summary	Present results and actionable insights.

3.1 Processing Flow

1. Load existing datasets.
2. Clean and structure the data using Pandas.
3. Preserve timestamps for time-based analysis.
4. Process Twitter for sentiment, trends and basic network analysis.
5. Process Instagram for content and engagement analysis.

6. Combine outputs into a simple visual dashboard.
7. Present insights that demonstrate the audience-intelligence concept.

4. Instagram Analytics

4.1 Time and Trend Analysis

Posting volume was grouped by hour. The analysis distinguishes posting frequency from performance: the hour with the most posts is not necessarily the hour with the best engagement.

The five highest average engagement-rate hours reported were 03:00, 08:00, 02:00, 17:00 and 20:00. The differences were small, so posting time should be treated as one feature among several.

Sunday, Tuesday and Friday were reported among the higher average-engagement days. Day of week should likewise be combined with content, format, account and audience signals.

- Average reach was calculated by posting hour.
- Average followers gained was calculated by hour.
- Daily posting volume, engagement rate and reach were analysed.
- Daily total shares were calculated as a propagation signal.

4.2 Content Category Analysis

Category	Posts
Photography	3,035
Fashion	3,034
Technology	3,025
Lifestyle	3,017
Food	3,010
Fitness	3,004
Music	3,003
Travel	2,968
Beauty	2,953
Comedy	2,950

The highest reported average engagement-rate categories were Music (0.042808), Fitness (0.042720) and Fashion (0.042615). The differences were small.

4.3 Hashtags and CTA

- Hashtag count did not show a simple guaranteed increase in engagement; the relationship was noisy.
- No CTA: average engagement rate 0.042232.
- CTA: average engagement rate 0.041875.
- The CTA result is an association only and does not prove that adding a CTA reduces engagement.

5. Twitter Conversation Analysis

5.1 Timeline Analysis

The conversation was divided into five-minute windows and separated into Original Tweets and Retweets. Original tweets represent new content creation, while retweets represent content sharing and amplification.

Key Finding

Retweets dominated the conversation. The reported retweet ratio was approximately 0.90, meaning roughly 90% of the observed conversation consisted of retweets.

- Original tweets remained comparatively low and steady.
- Retweets were substantially higher and more fluctuating.
- The conversation was strongly sharing-driven.
- The pattern indicates high content amplification and potentially high reach through repeated sharing.

5.2 VADER Sentiment Analysis

VADER was used as the baseline sentiment model. It produces negative, neutral and positive intensity scores and a compound score ranging from −1 to +1.

Compound Score	Class
≥ +0.05	Positive
≤ −0.05	Negative
−0.05 to +0.05	Neutral

5.3 Sentiment Results

View	Positive	Neutral	Negative
All Tweets	52.94%	38.42%	8.64%
Original Tweets Only	38.96%	46.56%	14.48%

- Overall conversation sentiment was net-positive.
- Positive content was partly amplified through retweets.
- Original tweets had higher neutral and negative proportions.
- Negative sentiment was more concentrated in original tweets.

6. Machine Learning – Sentiment Classification

6.1 Data Preparation and Leakage Prevention

- Training dataset: approximately 74K tweets initially; 57,178 usable after cleaning.
- Testing dataset: approximately 1K initially; 827 after cleaning.
- Final evaluation set: 398 genuinely unseen tweets.
- Approximately 52% of the provided test tweets initially overlapped with training data.
- Overlapping tweets were removed to create a fairer evaluation.

6.2 TF-IDF + Logistic Regression

8. Clean tweet datasets.
9. Remove irrelevant metadata where appropriate.
10. Check duplicates and train/test overlap.
11. Convert tweet text into numerical features using TF-IDF.
12. Train Logistic Regression.
13. Tune hyperparameters.
14. Evaluate only on genuinely unseen tweets.

TF-IDF (Term Frequency–Inverse Document Frequency) gives greater importance to words that help distinguish documents while reducing the influence of words that occur very frequently. Logistic Regression then learns patterns in those features to classify sentiment.

6.3 Model Results

Metric	Result
Initial accuracy	92.96%
Final accuracy after tuning	98.74%
Final evaluation set	398 unseen tweets
Misclassified	5 of 398

Remaining errors were mainly described as ambiguous or context-dependent tweets, including news statements, neutral tweets containing positive words, or tweets requiring domain knowledge.

7. Risk Prioritization Prototype

The current risk pipeline is: Time/Trend Analysis → Sentiment → Engagement Signals → Risk Score. The purpose is to prioritize posts for further investigation, not to declare confirmed threats.

7.1 Multiple-Signal Approach

- Sentiment indicates what the text expresses.
- Trend analysis identifies changes in activity.
- Shares and reach indicate content spread.
- Engagement provides an additional signal.
- Multiple signals are combined for preliminary prioritization.

7.2 Risk-Related Sentiment

Sentiment	Posts
Positive	13,804
Negative	8,547
Neutral	7,648

7.3 High-Share Negative Posts

Negative posts above the 90th percentile of shares represent approximately the top 10% of negative posts by shares. These are candidates for review, not confirmed threats.

Example Signal	Value
Shares	516
Reach	46,738
Engagement Rate	0.2398
Sentiment Score	−0.8655

7.4 Risk Score Formula

Formula

Risk Score = [(0.30 × Negative Score) + (0.25 × Share Score) + (0.25 × Reach Score) + (0.20 × Engagement Score)] × 100

The component scores are normalized to a 0–1 scale. The weights are manually selected heuristics. For example, 0.8, 0.7, 0.9 and 0.6 produce a combined score of 0.76, giving a Risk Score of 76.

Score	Level	Meaning
0–39	Low	Lower review priority.
40–69	Medium	Moderate review priority.
70–100	High	Higher review priority; not a confirmed threat.

8. Current Project Status

Component	Status
Instagram data validation	Completed
Time and trend analysis	Completed
Content analysis	Completed
Visualization work	Completed
Twitter sentiment datasets cleaned	Completed
Train/test leakage checks	Completed
TF-IDF + Logistic Regression	Built
Hyperparameter tuning	Completed
Final unseen-data evaluation	Completed
Rule-based risk score	Completed

9. Limitations and Data Gaps

- The prototype uses historical CSV files rather than live social-media APIs.
- No continuous streaming pipeline is implemented at this stage.
- Raw Instagram caption text is not available, so direct Instagram NLP cannot yet be performed.
- Account-to-account relationship data is missing, limiting network and propagation analysis.
- Demographic attributes are unavailable for audience profiling.
- Observed associations do not prove causation.
- performance_bucket_label must be checked for target leakage before predictive ML use.
- If engagement_rate is derived from the target/outcome, it should not be used as a predictor of that target.
- Likes, comments, shares and saves may be post-outcome variables and can cause leakage depending on the prediction point.
- The current risk score is a heuristic and is not a probability of a confirmed threat.

10. Recommended Next Technical Steps

15. Correlation Analysis – measure relationships among reach, impressions, likes, comments, shares, saves, followers gained and engagement rate.
16. Feature Engineering – build meaningful rates, ratios, time features and content features.
17. Machine Learning – define the prediction point and target, check leakage, then train and evaluate suitable models.
18. NLP – add raw captions and hashtags for sentiment and topic detection.
19. Propagation Analysis – add account and relationship data to study information flow.
20. Dashboard – combine trends, anomalies, topics, sentiment and risk signals in one human-readable interface.

21. Actual ML Threat Detection – create expert-reviewed labels, build NLP/propagation/network features, train and validate a supervised model, and connect predictions to the dashboard with human review.

11. Integrated System Flow

End-to-End Flow

Social-media data → Time/Trend Analysis → Sentiment Analysis → Engagement & Propagation Signals → Risk Score → Risk Level → Human Review

The intended framework treats sentiment, propagation, reach and engagement as complementary signals. A negative or high-engagement post is not automatically a threat.

12. Conclusion

The project has progressed from a raw-data proof of concept to a structured social-media analytics prototype. Instagram analysis provides a foundation for time, trend, content, engagement, reach, sharing and growth analysis. Twitter analysis demonstrates conversation dynamics and sentiment classification.

The machine-learning component adds a TF-IDF + Logistic Regression sentiment pipeline with explicit attention to train/test overlap and leakage prevention. The risk-prioritization component combines multiple signals using manually selected weights to support review prioritization.

Future development should connect these components through correlation analysis, feature engineering, richer NLP, propagation and network analysis, validated ML tasks, and an explainable dashboard.

Appendix – Key Findings

- 29,999 Instagram posts and 23 columns were analysed; zero missing values were reported in the checked dataset.
- Instagram engagement differences across leading categories were small.
- Twitter retweets accounted for approximately 90% of observed conversation activity.
- Overall Twitter sentiment was 52.94% positive, 38.42% neutral and 8.64% negative.
- Original tweets had a higher negative share of 14.48%.
- The tuned Logistic Regression sentiment model achieved 98.74% accuracy on 398 unseen tweets, with 5 misclassified.
- The risk prototype uses negative sentiment, shares, reach and engagement as weighted signals.
- High risk means higher review priority, not a confirmed threat.

SOURCES : ..\Introduction - PS & Approach.pdf
..\instagram.ipynb.pdf
..\Instagram_Social_Media_Analytics_Report.pdf
..\Meeting 2 - Twitter Dataset Analysis (EDA).pdf
..\Twitter Dataset Report.pdf
(time_and_trends)analysis.pdf
Instagram_Social_Media_Analytics_SIH_Presentation_v2.pptx
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